

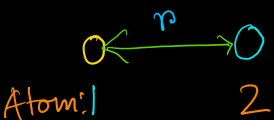
Harmonic Oscillator / Chapter 5 → McQuarrie, physical Chem.

Model for Vibrational Spectroscopy

Rigid Rotator : Model for Rotational Spectroscopy.

Simple Harmonic Oscillator: classical

want a model for $V(r)$ $0 < r \leq \infty$



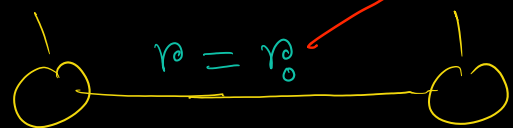
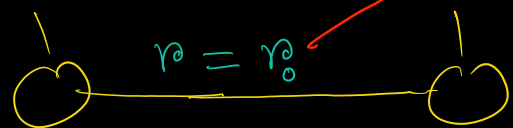
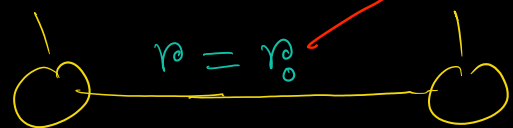
High energy → Dissociation → $V(\infty) = 0$

Condensed phase → Resists compression → $V(0) = \infty$

Gas phase → Finite size molecule → $\frac{dV(r_0)}{dr} = 0$

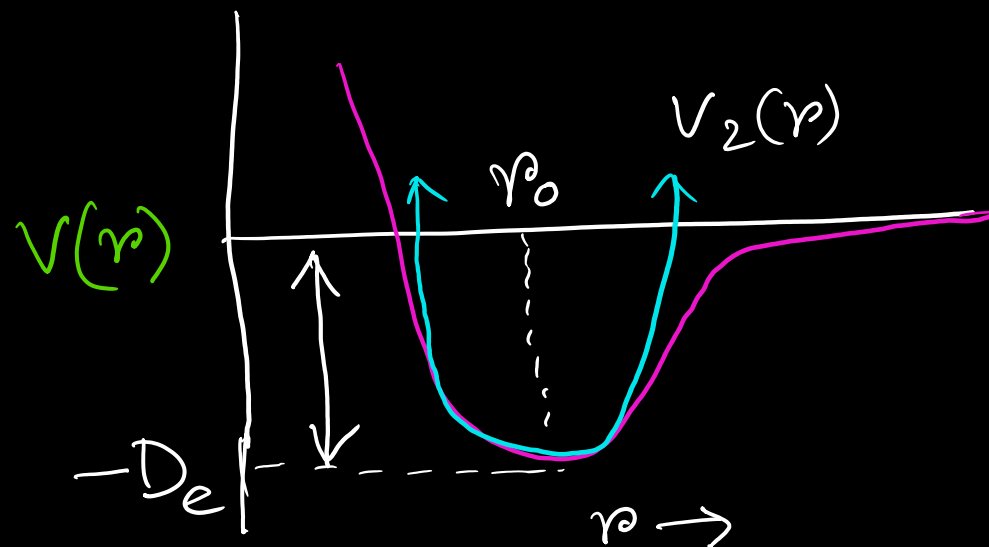
distance after compression

Equilibrium distance



$$F(r) = - \frac{dV(r)}{dr}$$

This (-)ve sign is because force and potential energy are at opposite direction.



$V(r)$ → Complicated

$V_2(r)$ → quadratic polynomial

$$V_2(r) = -D_e + \frac{1}{2} k (r - r_0)^2$$

$$x = r - r_0$$

k = spring constant

$$V(x) = V_2(r) + D_e$$

$$V(x) = \frac{1}{2} k x^2$$

Classical harmonic Oscillator Trajectories and Energy



Atom 1

Atom 2

$$V(x) = -\frac{dV(x)}{dx} = -\frac{d}{dx}\left(\frac{1}{2}kx^2\right) = -kx$$

Force linearly increases with displacement

$$F = ma \rightarrow -kx = m \frac{d^2x}{dt^2}$$

$$\text{or, } \frac{d^2x}{dt^2} = -\frac{k}{m}x, \quad \omega \equiv \sqrt{\frac{k}{m}}$$

$$\Rightarrow x(t) = A \cos(\omega t) + B \sin(\omega t)$$

At initial condition $B = 0$

$$x(t) = A \cos(\omega t)$$

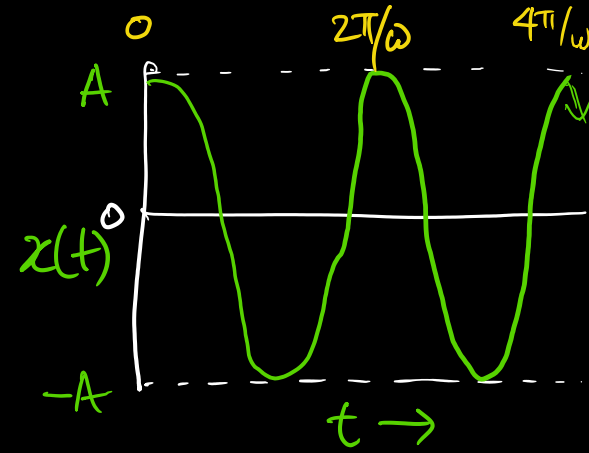
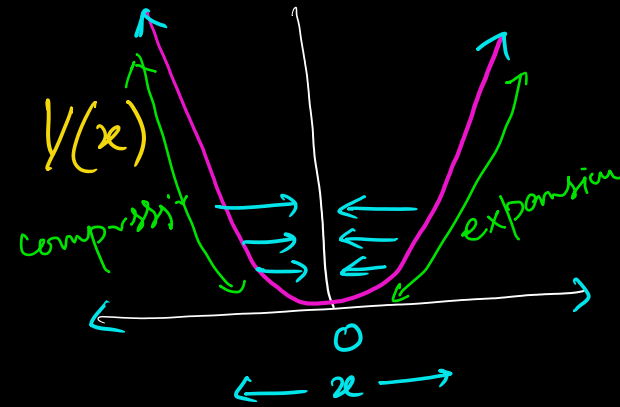
A = Amplitude

$$-A \leq x(t) \leq A$$

ω = Angular velocity

Solution

Try at home.



Energy

$$E(t) = \frac{1}{2} k A^2$$

k = force constant

Small $\rightarrow$ large $\rightarrow$ Stiff spring
loose spring

Harmonic oscillator wave function

$$\hat{H}\psi_n(x) = E_n \psi_n(x) \quad V(x) = \frac{1}{2} kx^2$$

$$E_n = \left(n + \frac{1}{2}\right) h\nu \quad n = 0, 1, 2, \dots \quad \nu = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}}$$

Frequency of oscillation

$$X = \gamma - \gamma_0$$

$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$

Reduced mass

$$\psi_n(x) = N_n \underbrace{H_n(\underbrace{\sqrt{\alpha} x})}_{\text{Hermite polynomials}} e^{-\alpha x^2/2} \quad \alpha = \frac{\sqrt{k\mu}}{\hbar}$$

Normalization Constant

Hermit
polynomial

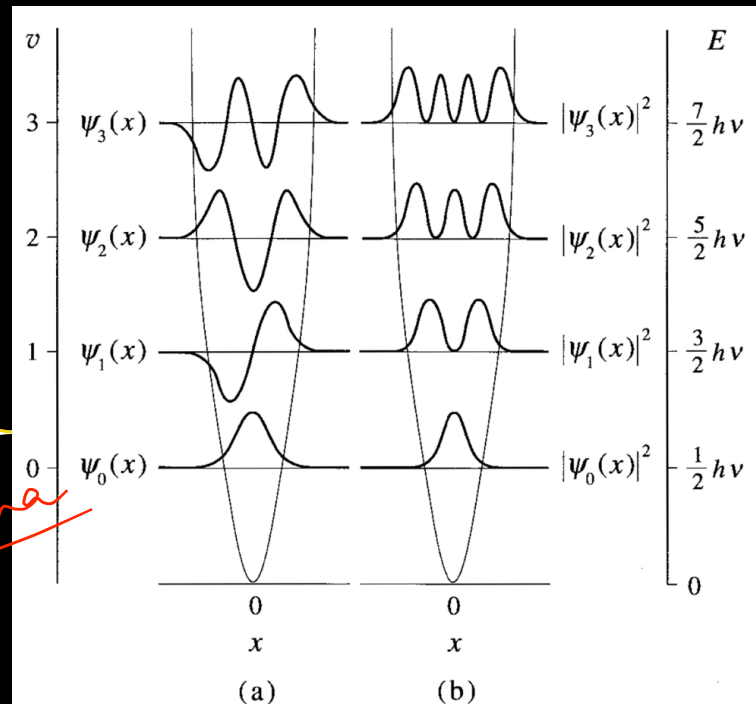
Gaussian Function

$$N_n = \frac{1}{\sqrt{2^n n!}} \left(\frac{\alpha}{\pi} \right)^{1/4}$$

$$H_n(\xi) = (-1)^n e^{\xi^2} \frac{d^n}{d\xi^n} e^{-\xi^2} \quad \xi = \sqrt{\alpha} x = \left(\frac{k\mu}{\hbar^2} \right)^{1/4} x$$

Small x
large x

FWTm
= Full width maximum



$$H_0(\Sigma) = 1$$

$$H_1(\mathbb{C}) = 2\mathbb{Z}$$

$$H_2(\xi) = 4\xi^2 - 2$$

$$H_3(\tau_9) = 8\tau_9^3 - 12\tau_9$$

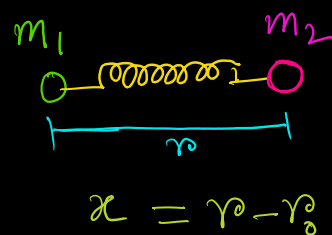
$$H_4(\mathcal{C}) = 16\mathcal{C}_4^4 - 48\mathcal{C}_3^2 + 12$$

Energy level of Quantum-Mechanical Harmonic Oscillator

$$\hat{H}\psi_n(x) = E_n\psi_n(x)$$

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x)$$

$$V(x) = \frac{1}{2}kx^2, \quad k \rightarrow \text{Spring Constant}$$

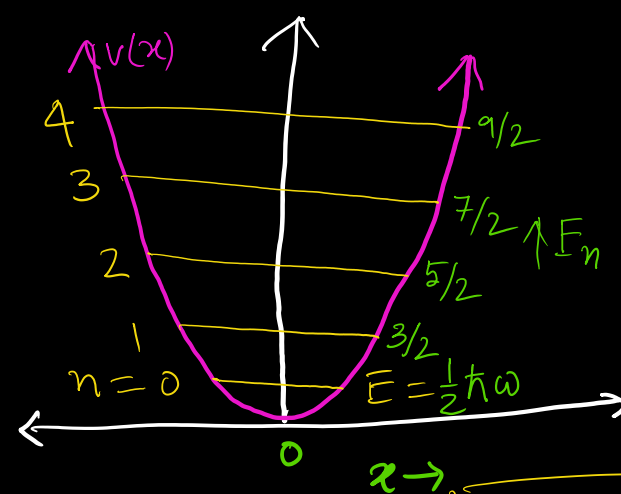


$$m = \mu = \frac{m_1 m_2}{m_1 + m_2} \quad \text{"Reduced Mass"}$$

$$\frac{d^2\psi_n(x)}{dx^2} = -\frac{2\mu}{\hbar^2} \left(E_n - \frac{1}{2}kx^2 \right) \psi_n(x)$$

$$E_n = \left(n + \frac{1}{2} \right) \hbar \nu / 2\pi = \frac{1}{2} \hbar \sqrt{\frac{k}{\mu}} \quad n = 0, 1, 2, 3, \dots$$

$$E_n = \left(n + \frac{1}{2} \right) \hbar \omega \quad \omega = \sqrt{\frac{k}{\mu}}$$



Ground state energy, $E_0 = \frac{1}{2}\hbar\omega$

Zero-point energy

3D Harmonic Oscillator

$$E_{n_x n_y n_z} = E_{n_x} + E_{n_y} + E_{n_z} \checkmark$$

$$= \left(n_x + \frac{1}{2}\right) h\nu + \left(n_y + \frac{1}{2}\right) h\nu + \left(n_z + \frac{1}{2}\right) h\nu$$

$$= \left(n_x + n_y + n_z + \frac{3}{2}\right) h\nu$$

$$V(x, y, z) = \underbrace{\frac{1}{2} k_x x^2}_{V_x} + \underbrace{\frac{1}{2} k_y y^2}_{V_y} + \underbrace{\frac{1}{2} k_z z^2}_{V_z}$$

$$\left[-\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) + V(x, y, z) \right] \psi(x, y, z) = E \psi(x, y, z)$$

$$\hat{H} = \hat{H}_x + \hat{H}_y + \hat{H}_z \quad \text{"Separable"}$$

$$H_i = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial i^2} + V_i(i) \quad i \in \{x, y, z\}$$

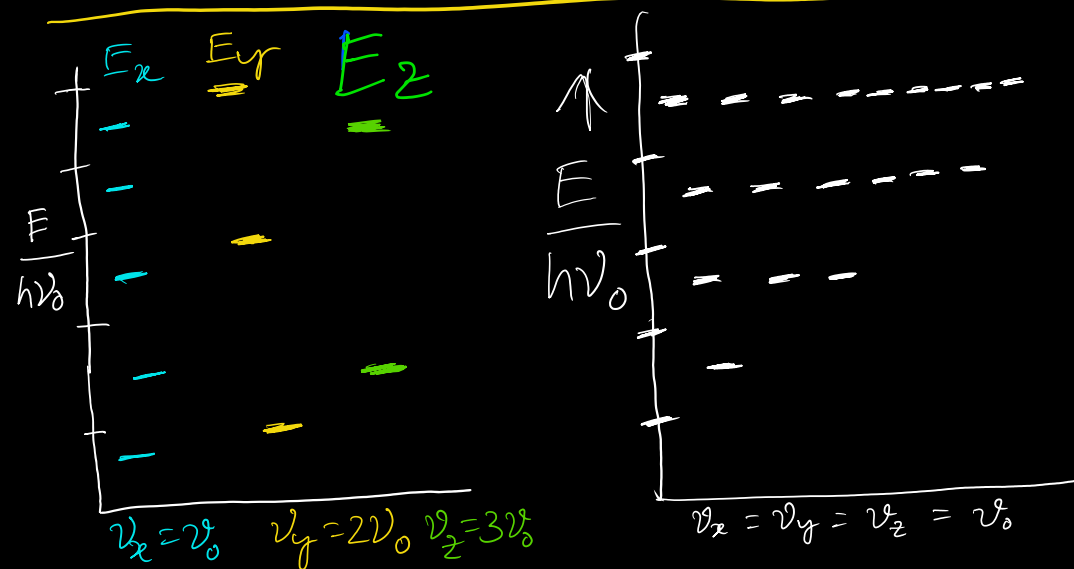
$$\hat{H}_i \psi_{n_i}(i) = E_{n_i} \psi_{n_i}(i)$$

$$E_{n_i} = h\nu_i \left(n_i + \frac{1}{2} \right) \quad \nu_i = \frac{1}{2\pi} \sqrt{\frac{k_i}{m}}$$

$$E_{n_x n_y n_z} = E_{n_x} + E_{n_y} + E_{n_z}; n_x, n_y, n_z = 0, 1, 2, \dots$$

$$\psi_{n_x n_y n_z}(x, y, z) = \psi_{n_x}(x) \psi_{n_y}(y) \psi_{n_z}(z)$$

Degeneracy in 3D simple harmonic oscillator



If $k_x = k_y = k_z = k$, then

$$E = \left(n_x + n_y + n_z + \frac{3}{2} \right) h\nu$$

$$n_x = 0, n_y = 0, n_z = 0$$

Symmetry $\Rightarrow$ Degeneracy

Rigid rotor model - 3rd Q.M. System

Model for rotating diatomic molecule.

Rotation frequency = ν_{rot}

Velocity of two masses are, $v_1 = 2\pi r_1 \nu_{\text{rot}}$ & $v_2 = 2\pi r_2 \nu_{\text{rot}}$

$$\Downarrow$$

$$v_1 = r_1 \omega$$

$$\Downarrow$$

$$v_2 = r_2 \omega$$

$$\omega = 2\pi \nu_{\text{rot}}$$

$\uparrow$
radian/sec or Angular speed

Kinetic energy of rigid rotor $\Rightarrow$

$$K = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

$$= \frac{1}{2} (m_1 r_1^2 + m_2 r_2^2) \omega^2$$

$$= \frac{1}{2} I \omega^2 \quad \left[\begin{array}{l} I = \text{moment of Inertia} \\ = m_1 r_1^2 + m_2 r_2^2 \end{array} \right]$$

Location of center of mass is $\Rightarrow m_1 r_1 = m_2 r_2$

$$\text{So, } I = \mu r^2$$

$\rightarrow$ reduced mass

Angular momentum, $L = I \omega$

Kinetic Energy, $K = \frac{L^2}{2I}$

Rigid rotor energy level

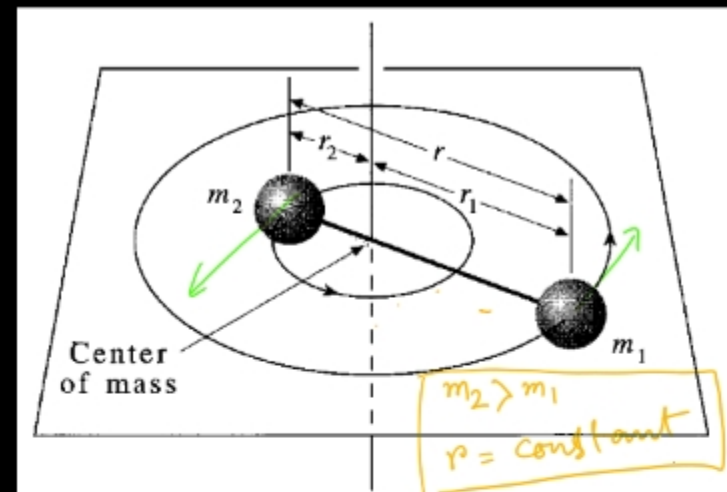
$V = 0$, no external force.

$$p = m v$$

$$E = \frac{p^2}{2m}$$

$$\hat{L}^2 = -\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \left(\frac{\partial^2}{\partial \phi^2} \right) \right]$$

θ, ϕ = Angle of rotation.



$$\hat{H} = \frac{\hat{L}^2}{2I}, \quad \hat{H} \Psi(\theta, \phi) = E \Psi(\theta, \phi)$$

$$E_J = \frac{\hbar^2}{2I} J(J+1), \quad J = 0, 1, 2, \dots$$

= Quantum number

$$B = \frac{h}{8\pi^2 I} \text{ (H}_2\text{)} \quad \text{Rotational Constant}$$

$$\tilde{B} = \frac{h}{8\pi^2 c I} \text{ (cm}^{-1}\text{)} \quad c = h\nu, \quad \nu = \frac{c}{\lambda}, \quad \tilde{\omega} = \frac{1}{\lambda}$$

$$E_T = \frac{\hbar^2}{2I} J(J+1) = h B J(J+1) = h c \tilde{B} J(J+1)$$

$$g_J = 2J + 1 \quad \text{Degeneracy}$$

Ref. Q.N.

Rigid Rotator is a model for a rotating diatomic molecule

Selection rule for rigid rotator says $\Rightarrow$

$$\Delta J = \pm 1$$

and

molecule must need permanent dipole moment to absorb electromagnetic radiation.

HCl has rotational spectrum

N₂

In case of absorption, molecule goes from $J \rightarrow J+1$

$$\begin{aligned} \text{So, } \Delta E &= E_{J+1} - E_J = \frac{\hbar^2}{2I} [(J+1)(J+2) - J(J+1)] \\ &= \frac{\hbar^2}{I} (J+1) = \frac{h^2}{4\pi^2 I} (J+1) \end{aligned}$$

Frequency: $\Delta E = h\nu$

$$\text{So, } \nu = \frac{h}{4\pi^2 I} (J+1), \quad J = 0, 1, 2, 3, \dots$$

$$\boxed{\nu = 2B(J+1)} \quad J = 0, 1, 2, \dots$$

where $B = \frac{h}{8\pi^2 I}$
 ↑
 Rotational Constant

$$\tilde{\nu} = \frac{\nu}{c}$$

$$\text{then, } \boxed{\tilde{\nu} = 2\tilde{B}(J+1)} \quad J = 0, 1, 2, \dots$$

$\tilde{B}$ is rotational constant expressed in units of wave no. (cm⁻¹)

$$\boxed{\tilde{B} = \frac{h}{8\pi^2 c I}}$$

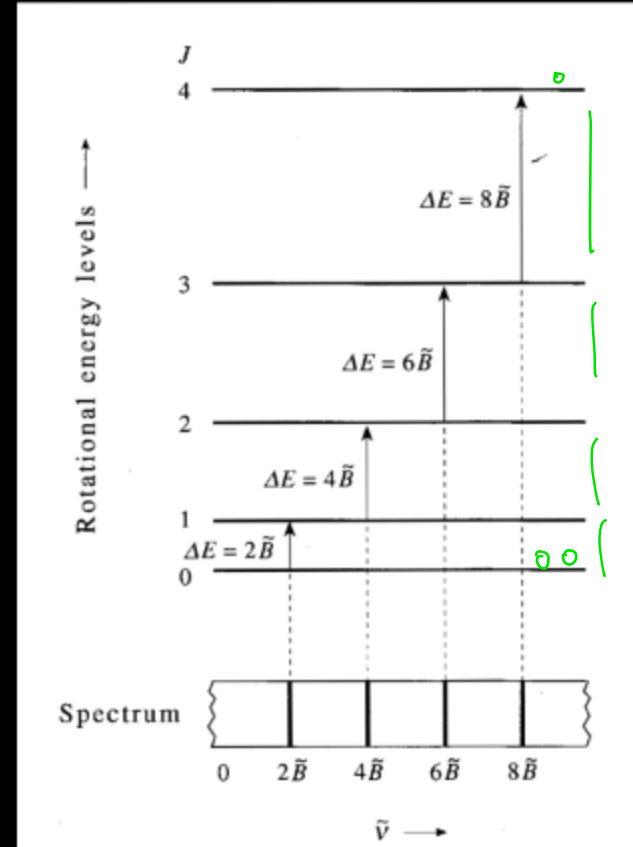


Diagram showing the moment of inertia I for a diatomic molecule with masses m_H and m_{Cl} separated by a distance r .

$$I = m_H r_1^2 + m_{Cl} r_2^2$$

where $r_1 + r_2 = r$ and $m_H r_1 = m_{Cl} r_2$.

Rigid Rotor Wave function

$$\hat{H} \Psi_{Jm}(\theta, \phi) = E_J \Psi_{Jm}(\theta, \phi)$$

Angular coordinates (polar + azimuthal)
J & m = Q, N

$$E_J = \frac{\hbar^2}{2I} J(J+1) \quad \hat{H} = -\frac{\hat{L}^2}{2I}$$

L = Angular Momentum
I = moment of inertia

$$\hat{L}^2 = -\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right]$$

$$\Psi_{Jm}(\theta, \phi) = \Theta_{Jm}(\theta) \Phi(\phi)$$

$$\frac{\sin \theta}{\Theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \frac{2IE}{\hbar^2} \sin^2 \theta = \frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} = -m^2$$

$$\Phi(\phi + 2\pi) = \Phi(\phi)$$

$$\Phi_m(\phi) = \frac{1}{\sqrt{2\pi}} e^{im\phi} \quad m = 0, \pm 1, \pm 2, \dots \quad -J \leq m \leq J$$

$$\Theta_{Jm}(\theta) = \sqrt{\frac{2J+1}{2} \frac{(J-|m|)!}{(J+|m|)!}} \underbrace{P_J^{|m|}(\cos \theta)}_{\text{Associated Legendre Polynomials}}$$

$$P_l^m(x) = \frac{(-1)^m}{2^l l!} (1-x^2)^{m/2} \frac{d^{l+m}}{dx^{l+m}} (x^2-1)^l$$

$$\Psi_{Jm}(\theta, \phi) = Y_J^m(\theta, \phi) = \underbrace{\sqrt{\frac{2J+1}{4\pi} \frac{(J-|m|)!}{(J+|m|)!}}}_{\text{normalization}} \underbrace{P_J^{|m|}(\cos \theta)}_{\text{polynomials}} \underbrace{e^{im\phi}}_{\text{Complex exponential}}$$

$$Y_0^0 = \frac{1}{\sqrt{4\pi}}$$

$$Y_1^0 = \sqrt{\frac{3}{4\pi}} \cos \theta$$

$$Y_1^{\pm 1} = \sqrt{\frac{3}{8\pi}} \sin \theta e^{\pm i\phi}$$

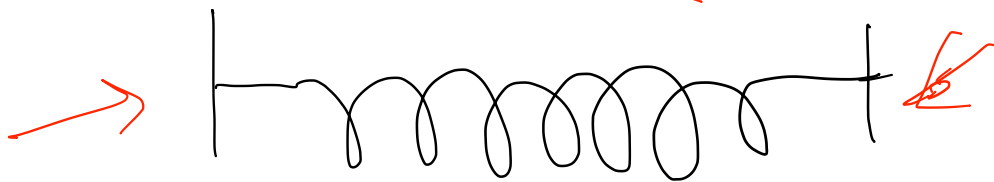
$$Y_2^0 = \sqrt{\frac{5}{16\pi}} (3\cos^2 \theta - 1)$$

$$Y_2^{\pm 1} = \sqrt{\frac{15}{8\pi}} \sin \theta \cos \theta e^{\pm i\phi}$$

$$Y_2^{\pm 2} = \sqrt{\frac{15}{32\pi}} \sin^2 \theta e^{\pm 2i\phi}$$

$$Y_l^m = \text{Spherical Harmonics}$$

Spring force Spring Spring force



your force

your force

